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- 1. What real-world decision or judgment is this model supposed to inform?
E.g. stock selection, portfolio selection (would consider risk balance as well), portfolio optimization**

The model supposed to inform long-term stock selection decisions by identifying companies that are likely to outperform a benchmark index over a defined investment horizon, based on fundamental financial indicators. The output can be used to construct a fundamentally driven equity portfolio.

- 2. What happens if the model is wrong? In which direction is error worse? Do we care more about false positives or false negatives?**

if the model is wrong there is possibility of loss in two direction - false positive and false negative. false positive is worse than false negative, in case of false negative there is opportunity loss but in case of false positive there will financial loss so we should care more about false positives.

- 3. What entity does a single prediction correspond to? E.g. A firm, A firm at a particular time, A firm in a time interval, etc.**

A firm at a particular time.

- 4. Is this unit naturally i.i.d., or is it dependent on others? E.g. Firms in same industry or a rolling window of metrics are not independent**

Data points are not strictly i.i.d., it dependent on firms in same industry.

- 5. Does one data point correspond to one real decision opportunity? (Sanity Check till now)**

Yes

- 6. At what exact time is the prediction made? End of a quarter? A year? A day (pls no)?**

predication made at the end of each financial reporting period (a quarter or a year)

7. Can you foresee any biases leaking into the feature space for this time period? E.g. Survivorship biased data or revised financials

Potential biases include survivorship bias, reporting delays, and revised financial statements. These are mitigated by using point-in-time data and including delisted firms where available.

8. What horizon is the outcome measured over? One year? 10 years? 17 quarters?

10 years

9. How is stock market return encoded into classes? 2 classes (above and below benchmark)? 4 classes? You could also classify based on 2 or more properties beyond just return (alpha wrt to index, rank or percentile of performance).

Stock returns are encoded into binary classes based on relative performance to a benchmark index over the prediction horizon. A stock is labeled positive if it generates positive alpha, i.e., outperforms the benchmark, and negative otherwise. This encoding directly aligns with the real-world objective of identifying stocks that beat the market, while remaining robust to noise in absolute return magnitudes.

10. Beyond just the financial features, is there other information that would be relevant? E.g. meta data like industry name, industry data like size and growth of industry, etc. If yes how would it be quantified? (I will try my best to acquire this data, cannot promise anything though)

Yes, beyond firm-level financial metrics, additional contextual information such as industry classification, firm size, and industry growth indicators is relevant. These features are quantified using categorical encodings for industry membership, logarithmic scaling of market capitalization for firm size, and numerical industry-level growth or return metrics.

11. What is the best way to evaluate the model? Cross validation? Rolling windows? What would best represent deployment of the model? CV is not always the best answer (it might be here, just not by default).

The model is evaluated using rolling-window (walk-forward) validation, which best reflects real-world deployment

12. What baselines or benchmarks would you like to beat? Mean prediction? Industry Average? Inflation? Index?

Model performance is benchmarked against market index returns, industry-average performance.

13. What transformations are allowed and preserve meaning? Normalization?

Meaning-preserving transformations such as normalization, log-scaling, and winsorization are applied to handle scale differences and outliers while preserving economic interpretation. Transformations are applied consistently across training and testing data.

14. How is order in information represented? Time order that is? Adjacent intervals? Rolling intervals?

Order in information is represented explicitly through chronological time ordering. Features are constructed using rolling time windows, where only information available prior to the prediction time is used. The model is trained and evaluated on time-ordered data, ensuring that past data predicts future outcomes and preventing any look-ahead bias. This approach closely mirrors real-world deployment, where decisions are made sequentially over time.

15. What kind of mistakes are penalized the most?

The model penalizes false positives more heavily, as incorrectly identifying a stock as a strong performer poses a greater financial risk than missing a potential outperformer.

16. For the purposes of inference, which of these features are casually related with the predicted variable, which are merely correlations? E.g. does +ve growth in both industry and stock imply correlation or causation?

Some features, such as profitability ratios and leverage metrics, are plausibly causally related to long-term returns through business fundamentals. Others, such as industry-wide growth trends or market sentiment indicators, are treated as correlational signals rather than causal drivers.

17. Are some features static (geography, industry) and some dynamic (ROE)? How are they represented in your feature space?

The feature space includes both static features (e.g., industry classification, sector) and dynamic features (e.g., ROE, revenue growth, debt ratios). Static features are encoded using categorical encodings, while dynamic features are updated at each time step based on the latest available financial data.